

MoonBit Typographical Compiler

Architecture & Features

Slide 1: Introduction to moon-typst

moon-typst brings modern typographical typesetting to the MoonBit ecosystem.

- â€¢ **Zero-FFI Pure MoonBit**: 100% portable across Native , JS, and WebAssembly targets
- â€¢ **Composable Pipeline** Separate MoonBit modules handle parsing , layout , and rendering
- â€¢ **Multi-Target Rendering**: Seamless output to SVG, semantic HTML5, and standard PDF 1.4
- â€¢ **Rich Mathematical Layout** First-class support for fractions , square roots , matrices , and limits

Slide 2: Mathematical Capabilities

Typesetting complex scientific formulas with pixel-perfect alignment :

$$\frac{1}{2} \frac{d}{dt} (\alpha + \beta) = \gamma + \delta$$

Multivariate transformations with matrix layouts :

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} u \\ v \\ w \end{pmatrix}$$

State vectors in 3D projection :

Slide 3: Architectural Pipeline

The internal pipeline consists of four modular packages :

- â€¢ **core**: Abstract syntax tree , color representations , and length unit conversions
- â€¢ **diag**: Source span tracking , diagnostic error collection , and colorized terminal reports
- â€¢ **eval**: Lexical scoping , variable environments , and cascading style rules
- â€¢ **parser**: Fast recursive descent lexer and token stream analyzer
- â€¢ **layout**: Spatial bounding box calculations , text measurement , and table layout
- â€¢ **render**: SVG vector emitter , HTML5 document generator , and PDF 1.4 binary serializer

Slide 4: Getting Started

Compile any Typst document directly using the CLI :

```
# Render to vector SVG
typst-render presentation.typ -o presentation.svg

# Render to standard PDF 1.4
typst-render presentation.typ -p -o presentation.pdf

# Render to semantic HTML5
typst-render presentation.typ --html -o presentation.html
```

Empowering next-generation cloud and WebAssembly document authoring!